

**Machine Learning Prediction of Antimicrobial Susceptibility Categories (S/I/R)
from Quality-Controlled Hospital Microbiology Data**

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Abstract

Hospital microbiology data are frequently affected by missing values, duplicate records, inconsistent naming, and heterogeneous coding across information systems, all of which limit their suitability for machine-learning analysis. This study developed a machine-learning framework to predict antimicrobial susceptibility of microorganisms, classified as sensitive (S) versus non-sensitive (non-S; combining intermediate and resistant), using patient-context features after systematic data quality assessment.

An end-to-end pipeline was designed to evaluate data quality in two dimensions, completeness and consistency, followed by preprocessing to standardize organism names, specimen types, antibiotic names, and susceptibility-result fields using dictionary mapping, regular expressions, and fuzzy matching. Intrinsic-resistance rules were incorporated, records that could not be interpreted reliably were flagged as REVIEW and

excluded from supervised learning, and repeated isolates were handled using a study-defined 30-day first-isolate rule to reduce bias from duplicate results.

After all preprocessing and eligibility criteria were applied, 528,134 records remained for model development. Four tree-based machine-learning models—Decision Tree, Random Forest, XGBoost, and CatBoost—were trained to predict sensitive (S) versus non-sensitive (non-S) outcomes from microbiology and patient-context features. To mitigate class imbalance, CTGAN-based augmentation was applied to the training data to balance the binary classes.

Model performance was evaluated using accuracy, precision, recall, and F1-score under three experimental settings with sequential feature removal. Random Forest achieved the best performance in all settings, with the highest accuracy of 81.66% and F1-score of 0.8166 when organism, antibiotic, specimen, sex, age, and ward features were included. When organism information was removed, accuracy decreased to 72.33%, and when both organism and ward information were removed, accuracy further decreased to 67.04%.

These findings indicate that systematically quality-controlled hospital microbiology data can support robust machine-learning prediction of antimicrobial susceptibility and that organism and antibiotic variables contribute most strongly to predictive performance.

Keywords: data quality assessment; data preprocessing; machine learning; antimicrobial susceptibility testing; binary classification; antimicrobial susceptibility prediction; antimicrobial resistance; CTGAN; Random Forest.

1. Introduction

Culture and antimicrobial susceptibility testing (AST) data are fundamental for the treatment of infectious diseases, antimicrobial stewardship, and antimicrobial resistance (AMR) surveillance in hospitals. Cumulative antimicrobial susceptibility reports, or antibiograms, are widely used to summarize institutional susceptibility patterns and support empiric antimicrobial selection. The CLSI M39 guideline provides an important framework for the analysis and presentation of cumulative antimicrobial susceptibility test data so that such reports are reliable and clinically useful (Clinical and Laboratory Standards Institute [CLSI], 2022a).

In practice, however, hospital microbiology data are often affected by multiple data quality problems, including missing values, duplicate records, non-standard organism and antibiotic names, inconsistent specimen labels, and heterogeneous coding across information systems. These problems reduce not only the reliability of cumulative susceptibility reporting but also the readiness of the data for machine-learning analysis. Systems such as WHONET and BacLink were developed to facilitate microbiology data management and standardization, highlighting the importance of data quality assessment and preprocessing before advanced analysis (Stelling et al., 2007; WHONET, 2026).

In recent years, machine learning has increasingly been applied to antimicrobial resistance and susceptibility prediction. Reviews of the literature have shown that tree-based models, particularly Random Forest and gradient-boosted trees, often perform well with real-world healthcare data because they can capture nonlinear relationships and accommodate complex interactions among variables (Anahtar et al., 2021; Ardila et al., 2025; Sakagianni et al., 2023). Nevertheless, many AMR prediction studies still face

common limitations, including poor source-data quality, class imbalance, inconsistent preprocessing pipelines, and limited discussion of how categorical AST labels were derived and standardized (Ardila et al., 2025; Sakagianni et al., 2023).

Accordingly, this study aimed to develop a machine-learning framework for predicting antimicrobial susceptibility under a binary classification setting (S vs non-S) from hospital microbiology data after systematic data quality assessment and preprocessing. The specific objectives were: (1) to assess and improve the quality of hospital microbiology data in terms of completeness and consistency before model development; (2) to develop a preprocessing workflow that standardizes organism names, antibiotic names, specimen types, and susceptibility results into a machine-learning-ready format; (3) to develop and compare the predictive performance of multiple machine-learning models for binary antimicrobial susceptibility classification (S vs non-S); and (4) to examine the contribution of microbiology variables and patient-context variables to predictive performance.

2. Related Work

Relevant studies can be grouped into four major areas: (1) surveillance systems and cumulative antimicrobial susceptibility reporting, (2) standardization and integration of microbiology data, (3) machine-learning applications for antimicrobial susceptibility prediction, and (4) methodological gaps related to data quality, class imbalance, and real-world implementation.

In the area of surveillance and reporting, CLSI M39 provides an important framework for the analysis and presentation of cumulative antimicrobial susceptibility

test data to support clinical decision-making and antimicrobial stewardship (CLSI, 2022a). At the global level, the World Health Organization established the Global Antimicrobial Resistance and Use Surveillance System (GLASS) to standardize the collection, analysis, and sharing of AMR data (World Health Organization, 2022, 2023). In Thailand, the National Antimicrobial Resistance Surveillance Center of Thailand publishes annual antibiogram reports and surveillance summaries, underscoring the direct role of AST data in AMR monitoring and antibiotic stewardship (National Antimicrobial Resistance Surveillance Center of Thailand, 2024, 2025).

In the area of standardization and microbiology data integration, several studies have shown that the major challenge in hospital microbiology datasets is not simply the absence of data but also inconsistency of formats, nonuniform coding, and fragmentation across information systems. WHONET and BacLink were developed to support the capture, conversion, processing, and analysis of microbiology data from legacy systems, particularly for surveillance use (Stelling et al., 2007; WHONET, 2026). In the Thai context, previous work has demonstrated ETL-based processing and visualization pipelines for bacterial profiles and antibiotic susceptibility profiles from hospital data, emphasizing data cleaning, quality assessment, and bias reduction before further analysis (Aramrat & Boonma, 2023).

For machine-learning applications, previous studies have demonstrated that ML can support antimicrobial stewardship, resistance phenotype prediction, and clinical decision support. Personalized antibiogram approaches based on electronic health records have shown the ability to predict antibiotic susceptibility and support narrower-spectrum therapy while maintaining adequate coverage (Corbin et al., 2022). Multitask learning frameworks have also been proposed to predict individualized resistance probabilities

across multiple antimicrobial classes (Goto et al., 2026). Collectively, these studies suggest that real-world clinical and microbiology data can be used for clinically meaningful susceptibility prediction (Anahtar et al., 2021; Corbin et al., 2022; Goto et al., 2026).

Despite these advances, major challenges remain. Recent reviews have emphasized that model performance in AMR prediction depends not only on algorithm choice but also on the quality and consistency of the underlying data, the handling of class imbalance, and the rigor of evaluation procedures (Ardila et al., 2025; Sakagianni et al., 2023). These limitations motivate the present study, which directly focuses on the integration of data quality assessment, systematic preprocessing, and binary antimicrobial susceptibility prediction (S vs non-S) using real-world hospital microbiology data.

3. Materials and Methods

3.1. Study Design and Data Source

This retrospective study used microbiology and anonymized patient-context data from Chiang Mai University hospital, a tertiary hospital in Chiang Mai province, Thailand, collected from 8 September 2011 to 2 January 2020. A total of 1,097,459 inpatient culture records were initially obtained from the IPD Culture database. Data quality assessment was performed for key microbiology and patient-context variables, including organism name, antibiotic name, specimen type, susceptibility result, sex, age, and ward of admission.

The records then underwent sequential preprocessing to clean, standardize, and transform the data. To ensure data quality and clinical relevance, the **eligibility criteria**

for retaining records were defined as follows: (1) records must originate from inpatient (IPD) cultures and successfully merge with the patient-context database via admission numbers; (2) susceptibility outcomes must be available and subsequently transformed into a binary classification (Sensitive [S] vs non-sensitive [non-S], where non-S combines Intermediate [I] and Resistant [R]); (3) records must pass the data quality assessment, excluding any entries flagged as 'REVIEW' due to unresolvable non-standard entries or missing values; and (4) inclusion was restricted to the first isolate per patient, per organism, per antibiotic within a 30-day period (the 30-day first-isolate rule) to mitigate bias from repeated sampling. After all preprocessing steps and eligibility criteria were applied, a final cohort of **528,134 records** remained for model development.

3.2. Research Ethics

This study was reviewed and approved by the Human Research Ethics Committee, Faculty of Medicine, Chiang Mai University, Committee 5 (Research ID: 8889; Study Code: FAM-2565-08889). The present study is a secondary analysis of data derived from a previously approved research project. It involved retrospective hospital microbiology and clinical record data, with no direct patient contact or intervention. All data were de-identified prior to analysis. The requirement for individual informed consent was waived by the ethics committee because the study posed minimal risk and did not adversely affect the rights or welfare of the patients.

3.3. Data Quality Assessment

This study assessed data quality in two primary dimensions, namely completeness and consistency, because both dimensions directly affect data readiness for machine

learning. Let the full dataset be denoted by $D = \{x_1, x_2, \dots, x_N\}$, where N is the total number of records and x_{iv} is the value of variable v in record i .

The completeness of variable v was defined as:

$$\text{Completeness}(v) = \frac{1}{N} \sum_{i=1}^N I(x_{iv} \text{ is not missing}),$$

where $I(\cdot)$ is the indicator function.

The consistency of variable v relative to a predefined standard-value set S_v was defined as:

$$\text{Consistency}(v) = \frac{1}{N} \sum_{i=1}^N I(x_{iv} \in S_v).$$

These measures were used to identify fields with missing values, non-standard naming, or inconsistent data formats before the preprocessing stage.

3.4. Data Preprocessing

The preprocessing workflow was designed for microbiology data originating from multiple sources and containing inconsistent naming conventions. It consisted of five main steps.

3.4.1. Data Standardization

Let r denote a raw value, such as an organism name, antibiotic name, specimen type, or susceptibility-result string, and let $T = \{t_1, t_2, \dots, t_m\}$ denote the corresponding set of standardized terms. The standardization function $f(r)$ was defined as:

$$f(r) = \begin{cases} \text{dictionary match,} & \text{if } r \text{ matches a predefined mapping} \\ \text{regex-normalized match,} & \text{if } r \text{ matches a standard pattern} \\ \arg \max_{t \in T} \text{sim}(r, t), & \text{if } \max_{t \in T} \text{sim}(r, t) \geq \tau \\ \text{REVIEW,} & \text{otherwise,} \end{cases}$$

where $\text{sim}(r, t)$ is the similarity score between raw value r and standardized term t , and τ is the fuzzy-matching threshold. This procedure was applied to organism names, antibiotic names, specimen types, and susceptibility-result fields.

3.4.2. Intrinsic-Resistance Checking

Let o_i denote the standardized organism for record i , a_i the standardized antibiotic, y_i the recorded susceptibility category, and R_{int} the rule base of organism-antibiotic pairs with known intrinsic resistance. The intrinsic-resistance inconsistency flag was defined as:

$$\text{IRFlag}_i = \begin{cases} 1, & \text{if } (o_i, a_i) \in R_{\text{int}} \text{ and } y_i = S \\ 0, & \text{otherwise.} \end{cases}$$

If $\text{IRFlag}_i = 1$, the record was flagged for review.

3.4.3. REVIEW Flag Assignment

Let status_i denote the final preprocessing status of record i . A record was assigned REVIEW status if it met at least one of the following conditions: unresolved organism mapping, unresolved antibiotic mapping, unresolved specimen mapping, invalid or ambiguous susceptibility-result coding, or intrinsic-resistance inconsistency.

$$\text{status}_i = \begin{cases} \text{REVIEW,} & \text{if the record cannot be reliably interpreted} \\ \text{KEEP,} & \text{otherwise.} \end{cases}$$

The supervised-learning dataset before duplicate reduction was therefore defined as:

$$D_{\text{sup}} = \{i: \text{status}_i = \text{KEEP and } y_i \in \{S, I, R\}\}.$$

3.4.4. First-Isolate Rule for Duplicate Reduction

Let record i contain patient identifier p_i , organism o_i , specimen s_i , and test date d_i . The first-isolate indicator was defined as:

$$F_i = \begin{cases} 1, & \text{if there is no earlier retained record } j \text{ such that } p_j = p_i, o_j = o_i, s_j = s_i, \\ & \text{and } 0 < d_i - d_j \leq 30 \text{ days} \\ 0, & \text{otherwise.} \end{cases}$$

The deduplicated dataset was then defined as:

$$D_{\text{dedup}} = \{i \in D_{\text{sup}} : F_i = 1\}.$$

In this study, the first-isolate rule was operationally defined as retaining only the first record within 30 days for the same patient-organism-specimen combination to reduce bias from repeated cultures from the same patient.

3.4.5. Final Dataset for Modeling

The final analysis dataset was defined as:

$$D_{\text{final}} = D_{\text{dedup}}.$$

The explanatory feature vector for each record was:

$$X = [\text{organism}, \text{antibiotic}, \text{specimen}, \text{sex}, \text{age}, \text{ward}],$$

and The target variable was defined as a binary outcome:

Sensitive (S) versus non-Sensitive (non-S), where non-S combines Intermediate (I) and Resistant (R).

This transformation was applied prior to model training and was consistently used throughout all experiments.

The study used the final categorical AST interpretations (S/I/R) recorded by the hospital laboratory as the prediction target. Because the study period spanned 2011 to 2020, breakpoint criteria may have varied over time. The study did not retrospectively

recalculate AST categories using a single breakpoint version across the full study period; instead, it used the categorical interpretations recorded in the hospital dataset at the time of testing. For contextual reference, the study period broadly spans the CLSI M100 editions from M100-S21 (2011) to the 30th edition (2020) (CLSI, 2011, 2020). In this study, category I denotes Intermediate.

3.5. Handling Class Imbalance

Because the dataset was imbalanced under the binary classification setting (Sensitive [S] vs non-sensitive [non-S], where non-S combines Intermediate [I] and Resistant [R]), CTGAN (Conditional Tabular Generative Adversarial Network) was applied to generate synthetic samples for the minority class.

CTGAN was trained using only the training dataset and was not applied to the test set. Synthetic samples were generated for the minority class (non-S) until both classes (S and non-S) reached equal sample sizes, ensuring a fully balanced binary class distribution for model training.

This approach ensured balanced class distribution and reduced bias toward the majority class during model training.

3.6. Experimental Design

The dataset was split into training and test sets using an 80:20 ratio, with stratified sampling to preserve the class distribution of the binary outcome (S vs non-S).

No separate validation set was used; instead, internal model evaluation was conducted using cross-validation within the training set. Stratified sampling was applied to preserve the class distribution of the binary outcome (S vs non-S) in both sets.

CTGAN-based augmentation was applied only to the training set before model training and after data splitting to balance the classes. The test set was kept unchanged to reflect real-world data distribution.

To evaluate the contribution of feature groups, three experimental settings were defined:

- Setting 1: organism + antibiotic + specimen + sex + age + ward
- Setting 2: antibiotic + specimen + sex + age + ward
- Setting 3: antibiotic + specimen + sex + age

This design allows assessment of the impact of microbiological and patient-context features on predictive performance.

3.7. Machine-Learning Models

This study compared four tree-based machine-learning models: Decision Tree, Random Forest, XGBoost, and CatBoost. Let x_i denote the feature vector of sample i . The final class prediction of each model can be expressed in general form as:

$$\hat{y}_i = \arg \max_{c \in \{S, I, R\}} P(y_i = c | x_i).$$

3.7.1. Decision Tree

Decision Tree selects splits by maximizing impurity reduction. In this study, Gini impurity was used:

$$G(t) = 1 - \sum_{c \in \{S, I, R\}} p(c | t)^2,$$

where $p(c | t)$ is the proportion of class c at node t . The impurity reduction obtained by split sat node t was defined as:

$$\Delta G(s, t) = G(t) - \frac{N_L}{N_t} G(t_L) - \frac{N_R}{N_t} G(t_R),$$

where N_t is the number of samples at node t , and N_L and N_R are the sample counts in the left and right child nodes, respectively.

3.7.2. Random Forest

Random Forest is an ensemble of multiple decision trees. The final prediction is obtained by majority vote:

$$\hat{y}_i = \text{mode}\{h_1(x_i), h_2(x_i), \dots, h_B(x_i)\},$$

where $h_b(x_i)$ is the class predicted by tree b , and B is the number of trees. The class probability can be written as:

$$P(y_i = c | x_i) = \frac{1}{B} \sum_{b=1}^B I(h_b(x_i) = c).$$

3.7.3. XGBoost

XGBoost is based on gradient boosting, in which trees are added sequentially to reduce residual error:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in \mathcal{F},$$

where $\mathcal{F}$ is the space of regression trees. The objective function is:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k),$$

where $l(\cdot)$ is the training loss and $\Omega(\cdot)$ is the regularization term.

3.7.4. CatBoost

CatBoost is a gradient-boosting algorithm designed specifically for categorical features and based on ordered boosting:

$$\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + \eta f_m(x_i),$$

where f_m is the base learner at iteration m and η is the learning rate.

3.7.5. Model Configuration and Experimental Environment

Categorical variables, including standardized organism name, standardized antibiotic name, standardized specimen type, sex, and ward, were encoded using Label Encoding before being used in the Decision Tree, Random Forest, and XGBoost models. CatBoost handled categorical features directly without explicit numeric encoding.

Hyperparameters were not optimized using Grid Search or Random Search. Instead, manual tuning was applied. The Decision Tree model used `max_depth = 10`. Random Forest used `n_estimators = 100`. XGBoost used `n_estimators = 100`, `learning_rate = 0.1`, and `max_depth = 6`. CatBoost used `iterations = 100`, `learning_rate = 0.1`, and `depth = 6`. Macro F1-score was used as the primary model-selection criterion under the binary classification setting (S vs non-S).

To ensure reproducibility, a random seed of 42 was used in all steps involving randomness, including data splitting, shuffling after CTGAN-based augmentation, and model training for Decision Tree, Random Forest, XGBoost, and CatBoost.

All analyses were conducted using Python 3.12 with the following libraries: pandas 3.0.2, scikit-learn 1.8.0, xgboost 3.2.0, catboost 1.2.10, and ctgan 0.12.1.

3.8. Evaluation Metrics

Model performance was evaluated using accuracy, Precision, Recall, and F1-score, and macro F1-score. All evaluation metrics were computed under the binary classification setting (S vs non-S). Particular emphasis was placed on macro F1-score because the class distribution was imbalanced.

Accuracy was defined as:

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n I(\hat{y}_i = y_i).$$

For each class c , precision and recall were defined as:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}.$$

The F1-score for class c was defined as:

$$F1_c = \frac{2 \times \text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

The macro F1-score was computed as:

$$\text{Macro F1} = \frac{1}{3} \sum_{c \in \{S, I, R\}} F1_c.$$

4. Results

4.1. Data Quality Assessment and Preprocessing Results

Before model development, an initial set of 1,097,459 inpatient culture records was retrieved from the IPD Culture database. Data quality was assessed in terms of completeness and consistency for key microbiological and patient-related variables. The overall completeness of the dataset was high; antibiotic names, specimen types, and susceptibility outcomes demonstrated 100.00% completeness, while organism names showed 96.22% completeness.

Following the quality assessment, the data underwent systematic preprocessing, including dictionary mapping, regular expression-based cleaning, and fuzzy matching techniques to standardize terminology and resolve inconsistencies. During this process, records with ambiguous or unresolvable entries were flagged as 'REVIEW'. A total of 30,562 records (approximately 4.75% of the filtered dataset) were excluded, primarily due to unresolved organism nomenclature or conflicts related to intrinsic resistance rules. This step ensured high label reliability for subsequent supervised learning.

The eligibility criteria for inclusion in this study required that the data: (1) originated from inpatient (IPD) cultures and were successfully merged with the clinical IPD dataset using admission numbers; (2) contained valid antimicrobial susceptibility results; (3) passed quality control by excluding entries flagged as 'REVIEW' or containing missing standardized nomenclature; and (4) followed a 30-day first-isolate rule per patient, organism, and antibiotic to reduce bias from repeated sampling. After applying all preprocessing workflows and eligibility criteria, a total of 528,134 records remained for model development.

To align with clinical decision-making and model stability, the antimicrobial susceptibility outcome was subsequently reformulated into a binary classification problem. Specifically, “Sensitive (S)” was defined as the positive class, while “Intermediate (I)” and “Resistant (R)” were combined into a single “non-Sensitive (non-S)” class. This transformation facilitated clearer interpretation and improved model robustness in downstream analysis.

To further address class imbalance, CTGAN was applied to the binary outcome to generate synthetic samples for the minority class. The augmentation process ensured that both classes (S and non-S) were equally represented, resulting in a fully balanced dataset used for machine learning experiments.

4.2. Model Comparison Results

The comparison results of the four machine learning models across the three experimental settings are presented in Table 1. Model performance was evaluated using accuracy, Precision, Recall, and F1-score, and macro F1-score, ensuring balanced assessment across both classes in the binary classification setting.

As shown in Table 1, Random Forest consistently achieved the best performance across all experimental settings and evaluation metrics. In Setting 1, which utilized the full feature set (organism, antibiotic, specimen type, sex, age, and ward), Random Forest achieved the highest performance with an accuracy of 81.66% and a macro F1-score of 0.8166. XGBoost and CatBoost demonstrated competitive performance, while Decision Tree showed comparatively lower accuracy.

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367 In Setting 2, where organism information was excluded, model performance
368 decreased across all models. Random Forest remained the best-performing model,
369 achieving an accuracy of 72.33% and a macro F1-score of 0.7233. This reduction
370 highlights the importance of organism-level information in antimicrobial susceptibility
371 prediction.

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373 In Setting 3, which further removed ward information and retained only antibiotic,
374 specimen type, sex, and age, performance declined further. Random Forest again
375 achieved the highest performance among the models, with an accuracy of 67.04% and a
376 macro F1-score of 0.6704.

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378 Overall, model performance decreased consistently as key features were removed,
379 with the largest performance drop observed after excluding organism information. These
380 findings emphasize the critical role of microbiological variables, particularly organism
381 identity, in predicting antimicrobial susceptibility.

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Setting	Features	Model	Accuracy	Macro		
				Precision	Recall	F1
Setting 1	Org, Anti, Specimen, Sex, Age, Ward	Decision Tree	74.41%	0.7448	0.7441	0.7439
		Random Forest	81.66%	0.8167	0.8166	0.8166
		XGBoost	77.37%	0.7738	0.7737	0.7737
		CatBoost	75.83%	0.7583	0.7583	0.7583
Setting 2	Anti, Specimen, Sex, Age, Ward	Decision Tree	66.14%	0.6617	0.6614	0.6612
		Random Forest	72.33%	0.7233	0.7233	0.7233
		XGBoost	69.05%	0.6905	0.6905	0.6904
		CatBoost	67.74%	0.6775	0.6774	0.6773
Setting 3	Anti, Specimen, Sex, Age	Decision Tree	65.64%	0.6600	0.6600	0.6600
		Random Forest	67.04%	0.6700	0.6700	0.6700
		XGBoost	66.80%	0.6700	0.6700	0.6700
		CatBoost	65.93%	0.6600	0.6600	0.6600

Table 1. Prediction results for antimicrobial susceptibility

(Binary classification: S vs non-S) across the three experimental settings.

4.3. Effect of Feature Reduction

To evaluate the contribution of feature groups in predicting antimicrobial susceptibility (binary classification: Sensitive [S] vs non-sensitive [non-S]), the performance of the best model (Random Forest) was compared under progressively reduced feature sets, as shown in Table 2.

Setting	Features	Accuracy	Macro			Accuracy Difference
			Precision	Recall	F1	
1	Setting Org, Anti, Specimen, Sex, Age, Ward	81.66%	0.8167	0.8166	0.8166	—
2	Setting Anti, Specimen, Sex, Age, Ward	72.33%	0.7233	0.7233	0.7233	-9.33%
3	Setting Anti, Specimen, Sex, Age	67.04%	0.6704	0.6704	0.6704	-14.62%

Table 2. Comparison of Random Forest performance under sequential feature reduction (Binary classification: S vs non-S).

Model performance declined consistently as key features were removed. Excluding organism information led to a substantial decrease in accuracy from 81.66% to 72.33%. Further removal of contextual features resulted in an additional decline to 67.04%.

This trend highlights the importance of microbiological variables—particularly organism and antibiotic information—in predicting antimicrobial susceptibility. The results indicate that simplified feature sets are insufficient to fully capture the complexity of susceptibility patterns.

4.4. Class-Level Performance of the Best Model

To further examine model behavior, class-wise performance of the best model (Random Forest) was evaluated under the binary classification setting, as shown in Table 3.

Setting	Class	Precision	Recall	F1-score
Setting 1	S	0.82	0.82	0.82
	non-S	0.81	0.81	0.81
Setting 2	S	0.72	0.72	0.72
	non-S	0.72	0.72	0.72
Setting 3	S	0.67	0.67	0.67
	non-S	0.67	0.67	0.67

Table 3. Class-wise classification performance of Random Forest.

The model demonstrated balanced performance across both classes (S and non-S) in all feature settings. Performance was highest when the full feature set was used and declined gradually as features were removed.

Feature importance analysis indicated that antibiotic and organism variables contributed most strongly to model performance, followed by specimen type, while demographic variables such as sex and age had relatively smaller effects. These findings

suggest that microbiological characteristics are the primary drivers of antimicrobial susceptibility prediction.

4.5. Effect of CTGAN-Based Class Balancing

To address class imbalance, CTGAN was applied to the training dataset under the binary classification setting (S vs non-S). Before augmentation, the dataset showed an imbalance between the two classes, with the non-S class being underrepresented.

CTGAN was trained using only the training data, and synthetic samples were generated for the minority class (non-S) until both classes reached equal representation.

After augmentation, the training dataset became fully balanced, with equal proportions of S and non-S samples. This balanced distribution enabled the model to learn more effectively from both classes, reducing bias toward the majority class and improving overall predictive performance.

The model demonstrated behavior consistent with known clinical patterns of antimicrobial susceptibility. Antibiotics associated with high effectiveness showed higher predicted sensitivity rates, while those commonly associated with resistance exhibited lower predicted sensitivity.

These observations indicate that the model captures clinically meaningful relationships between pathogens and antimicrobial agents and supports its potential application in clinical decision-support systems.

5. Discussion

This study developed and evaluated a machine-learning framework for predicting antimicrobial susceptibility under a binary classification setting (Sensitive [S] vs non-sensitive [non-S], where non-S combines Intermediate [I] and Resistant [R]) using microbiology and patient-context data after systematic data quality assessment and preprocessing. The results demonstrate that high-quality, standardized clinical microbiology data can support robust predictive modeling, with Random Forest consistently achieving the best performance across all experimental settings.

In the model using the full set of features—including organism, antibiotic, specimen type, sex, age, and ward information—Random Forest achieved the highest performance, with an accuracy of 81.66% and a macro F1-score of 0.82. This strong performance reflects the ability of ensemble tree-based models to capture complex, non-linear relationships between microbiological characteristics and patient context. The balanced precision and recall values indicate that the model performs consistently across both classes (S and non-S), suggesting that the applied CTGAN-based augmentation effectively mitigated class imbalance without introducing substantial bias.

When microbiological and contextual features were sequentially removed, model performance declined consistently. Specifically, removing organism information resulted in a decrease in accuracy to 72.33%, and further removing additional contextual variables led to a further decline to 67.04%. This trend highlights the critical importance of microbiological variables, particularly organism and antibiotic information, in predicting antimicrobial susceptibility. The marked performance reduction after

excluding organism-level data indicates that pathogen-specific characteristics are key determinants of susceptibility patterns.

Feature importance analysis further supports these findings. Antibiotic and organism variables contributed most strongly to model performance, followed by specimen type, while demographic variables such as sex and age had comparatively smaller contributions. These findings are consistent with clinical knowledge, where antimicrobial susceptibility is primarily driven by pathogen identity and antimicrobial agent characteristics rather than patient demographics alone.

The use of CTGAN for data augmentation played a crucial role in addressing class imbalance. By generating synthetic samples for the minority class (non-S) until both classes were equally represented, the model was able to learn more effectively from underrepresented patterns. Importantly, CTGAN was applied only to the training dataset, ensuring that the test data remained representative of real-world distributions and reducing the risk of overfitting.

From a clinical perspective, the model demonstrated behavior consistent with real-world antimicrobial susceptibility patterns. Antibiotics known to have high effectiveness showed higher predicted sensitivity rates, whereas those associated with resistance exhibited lower sensitivity predictions. This indicates that the model is not only statistically robust but also clinically meaningful, supporting its potential application in decision-support systems for antimicrobial selection.

Despite these strengths, several limitations should be acknowledged. First, the study was conducted using data from a single institution, which may limit generalizability to other healthcare settings. Second, although CTGAN improves class

balance, synthetic data generation may introduce artifacts that do not fully reflect real-world variability. Third, the binary classification approach simplifies susceptibility outcomes by grouping intermediate and resistant categories together, which may reduce granularity for certain clinical applications.

Future work should include external validation using multi-center datasets to assess generalizability, as well as the integration of additional clinical variables such as comorbidities, prior antibiotic exposure, and laboratory parameters. Expanding the framework to more granular prediction tasks or incorporating temporal modeling may further enhance its clinical utility.

In conclusion, this study demonstrates that machine-learning models trained on quality-controlled microbiology data can effectively predict antimicrobial susceptibility under a binary classification framework. The findings highlight the importance of microbiological features—particularly organism and antibiotic information—and support the use of Random Forest as a robust model for this task. The proposed pipeline provides a scalable approach for leveraging hospital data to support antimicrobial stewardship and improve clinical decision-making.

5.1. Limitations

This study has several limitations. First, the data originated from a single tertiary-care hospital, which may limit generalizability to other institutions. Second, the retrospective design relied on routine hospital data collected over multiple years, and laboratory practices or breakpoint criteria may have varied during the study period. The study used the categorical S/I/R results recorded in the hospital database and did not retrospectively recalculate all across the full period. Third, CTGAN-based augmentation

was applied only to the training set before model training and after data splitting and may not fully capture the true structure of minority-class observations. Fourth, external validation was not performed, and the models were evaluated only within the source institution. Fifth, categorical variables were label-encoded for Decision Tree, Random Forest, and XGBoost for computational convenience; although tree-based methods are generally less sensitive to such encoding than linear models, arbitrary ordinal effects cannot be ruled out completely. Finally, the evaluation design used group-based splitting by patient ID to reduce leakage, but it did not constitute a fully temporal validation strategy for prospective deployment.

These limitations indicate that future work should include external validation across hospitals, temporal validation using later-year cohorts, and additional comparison of encoding strategies and class-imbalance methods. Future studies may also investigate model calibration, uncertainty estimation, and clinically interpretable deployment frameworks for decision support.

6. Conclusion

This study proposed a machine-learning framework for predicting antimicrobial susceptibility under a binary classification setting (Sensitive [S] vs non-sensitive [non-S]) using hospital microbiology data after systematic data quality assessment and preprocessing.

The framework integrated data quality assessment, standardization of organism, antibiotic, and specimen labels, exclusion of unreliable records, intrinsic-resistance

checking, and duplicate reduction using a 30-day first-isolate rule prior to model development.

The experimental results demonstrated that Random Forest achieved the best performance across all feature settings. Using the full feature set, the model achieved an accuracy of 81.66% and a macro F1-score of 0.8166. When organism information was removed, performance decreased to 72.33%, and further removal of contextual variables reduced performance to 67.04%.

These findings highlight the critical role of microbiological features, particularly organism and antibiotic information, in predicting antimicrobial susceptibility. The results indicate that high-quality, standardized hospital microbiology data can support robust machine-learning prediction and provide a foundation for future clinical decision-support systems.

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